Policy Title

Defining, Identifying, and Handling Outlier Data

Effective Date

March 1, 2025

Last Revised Date

N/A

Responsibilities

Personnel assigned to the Data Analytics Strategy Management (DASM) Division are the subject matter experts and points of contact for questions and concerns regarding the content and application of this policy. Modifications to this policy should be discussed by the Data Governance Committee (DGC) prior to implementation to ensure they follow department standards, policies, and procedures for how data should be designed, collected, developed, stored, utilized, and destroyed.

Scope and Audience

This policy applies to all analytical reporting originating within the Fire and Rescue Department (FRD).

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1. Policy Purpose

   In compliance with Standard Operating Procedure X this policy establishes how outlier data are defined, identified, and handled.

2. Policy Statement

   Standardizing the way in which outlier data are defined, identified, and handled mitigates the potential negative impact resulting from the erroneous or improper use of data in analysis including, but not limited to, producing misleading output and drawing inappropriate or inaccurate conclusions.

3. Known Issues Impacting Data Quality and Integrity

   Understanding how data is generated informs how outlier data are defined. Through this understanding, a methodical investigation and analysis to determine whether outlier data points are true or erroneous, or whether they capture a meaningful signal versus irrelevant noise, can be conducted.

   Documenting changes to the data generation process provides context when trends in the data shift, helping to understand whether a trend is due to actual change in an occurrence or change to a procedure or business process (e.g. extended call processing times during COVID were caused by the addition of screening questions, not by a change in call taker aptitude, call volume, or emergency type caused by the pandemic). Procedure or business process-based outliers are more easily detected when error-based outliers are removed from the data. Awareness of the impact caused by changes to procedure and business processes informs both future policy considerations and analyses forecasting outcomes based on historical data. Changes to procedural or business processes should be documented via Informational Bulletin for historical record keeping.

**Data Originating with the Department of Public Safety Communications (DPSC)**

Response time data typically originates with the call taker and dispatcher's handling of emergency and non-emergency calls from the public to the DPSC. Although the FRD is accountable for the agency's overall response time (call received to first unit arrived), the FRD does not have direct influence over call processing time (call received to first unit dispatched), which is the result of DPSC's internal business and operating policy and procedures. Known issues affecting data originating with DPSC are identified below.

a. On joint calls with the police department, response time analysis must consider the timestamp when the FRD event entry begins thus capturing the call processing for only the FRD's response. This is particularly critical in cases where the police department is dispatched first.
b. On joint calls in Fairfax County with automatic aid partner response, when an agency other than the FRD arrives first, that agency's response is slowed by a known delay in the CAD2CAD process of sending the request for aid, the agency acknowledging receipt of the request, then permitting the dispatch of their units. This processing delay is typically 30 seconds or less.
c. When call takers do not abandon an event screen at the conclusion of a call where no event was entered, the "pick-up" or call received time for the subsequent call will be recorded as the "pick-up" time for the previous call which was not properly abandoned. This creates an inaccurate record of the call processing time for the next legitimate call which cannot be programmatically identified or modified in the original data source and must be mitigated manually.
d. During computer-aided dispatch (CAD) downtime, status changes communicated verbally over the radio may be recorded erroneously in writing, reentered inaccurately due to typos, or not reentered at all due to human error. If identified within 90 days of the occurrence, it is possible to make a request of DPSC to correct the original data source (ICAD_Live) before it is permanently documented in ICAD_Archive, however DPSC is under no obligation to make the correction and, therefore, these occurrences must be mitigated manually.

**Unit-initiated Response Data**

Response time data continues to be created by unit personnel through voiced actions over the radio and input actions on the mobile computer-aided dispatch terminal (MCT). This data is vulnerable to both technical error and human error or malfeasance. Known issues effecting unit-initiated response time data are identified below.

a. Units may press enroute before the unit leaves the station. While this erroneously reduces the turnout time for the unit, it artificially increases the travel time to the scene.
b. Units may not press the on-scene when they arrive at an incident.
c. Units may not press the "with patient" button on their radio when they arrive at the patient's side.
d. Units may be unable to update their status if they are in a location with poor cellular connectivity.

**Discrepancies Between CAD and Records Management System (RMS) Data**

Best practice is for personnel to download CAD data to their fire and electronic patient care reports (ePCR) to document the response timestamps recorded in CAD. However, in cases where the response timestamps in CAD are missing or known to be inaccurate, personnel may manually enter or override the values downloaded by CAD. When response time discrepancies between CAD and the RMS exist, the conflict must be resolved to preserve the integrity of the data used for analysis.

4. Definitions
    a. Outliers: Data points significantly beyond the expected or normal range. [i] Unless specified otherwise, an outlier data point is one that is greater than or less than three standard deviations above or below the arithmetic mean of the dataset.
        i. True: Real data representing an extenuating circumstance or occurrence beyond the expected range.
        ii. Error-based: Data caused by human or technical errors which is not reflective of the actual occurrence or phenomenon.
    b. Call Processing
        i. Begins with the timestamp recorded when the call taker either begins typing into an event screen or presses F10 to obtain the Automatic Number Identification/Automatic Location Identification (ANI/ALI) of the caller's location, which populates the event screen with the caller's approximate location, for an FRD event type.
        ii. Ends with the timestamp recorded when the first unit from any CAD2CAD unit is dispatched.
        iii. High outliers are any events where the call processing time is longer than three standard deviations above the average call processing time during the preceding 90 days. Low outliers are the number of events at the low end of the distribution equal with the total number of high outliers, ordered by creation datetime. This represents 0.6 percent of the dataset; 0.3 percent at each end of the distribution.
            - Low outliers are removed from the call processing dataset because this helps normalize the distribution of the data. Normalizing the dataset makes statistics derived from it less vulnerable to the impact of wide variability and makes descriptions of central tendency (mean, median, mode), which assume a normal distribution, more reliable. Additionally, the lowest call processing values are typically indicative of a technical or human error as the minimum time to process a call is the length of the automated recording at the beginning of each emergency call which is approximately 12 seconds long.
    c. Turnout
        i. Begins with the timestamp recorded by FRD's CAD when the unit is dispatched.
            - On requests for mutual aid, there may be a discrepancy in the timestamps between the requesting agency's CAD and the FRD's CAD due to the CAD2CAD processing delay. For the purposes of internal analysis, the FRD's record will be considered the authoritative record.
        ii. Ends with the timestamp recorded when the unit either presses the enroute button on their mobile computer terminal (MCT) or when the dispatcher presses the enroute button at their terminal if the status is communicated verbally. The 'cterm' column in the CAD table un_hi will denote whether the unit's or the dispatcher's terminal recorded the timestamp.
        iii. Outliers are any calls with a turnout time longer than three standard deviations above the average turnout time during the preceding 90 days. This represents approximately 0.3 percent of the dataset.
            - Low outliers are not removed from the turnout time dataset because zero-second turnout times are not uncommon, particularly in cases where a unit is monitoring radio traffic and CAD queues in anticipation of being dispatched or self-dispatches on already dispatched events.
    d. Travel Time

i. Begins with the timestamp recorded by FRD's CAD when the unit either presses the enroute button on their MCT or when the dispatcher presses the enroute button at their terminal if the status is communicated verbally. The 'cterm' column in the CAD table un_hi will denote whether the unit's or the dispatcher's terminal recorded the timestamp.
    - On requests for mutual aid, there may be a discrepancy in the timestamps between the requesting agency's CAD and the FRD's CAD due to the CAD2CAD processing delay. For the purposes of internal analysis, the FRD's record will be considered the authoritative record.

ii. Ends with the timestamp recorded when the unit either presses the arrived button on their MCT or when the dispatcher presses the arrived button at their terminal if the status is communicated verbally. The 'cterm' column in the CAD table un_hi will denote whether the unit's or the dispatcher's terminal recorded the timestamp.
    - In the case where no arrived timestamp is recorded, the unit's "with patient" or transport timestamp may be used as a proxy to their arrival time.

iii. Outliers are any calls with a travel time longer than three standard deviations above the average travel time during the preceding 90 days. This represents approximately 0.3 percent of the dataset.

e. Response Time

i. Begins with the timestamp recorded when the call taker either begins typing into an event screen or presses F10 to obtain the Automatic Number Identification/Automatic Location Identification (ANI/ALI) of the caller's location, which populates the event screen with the caller's approximate location, for an FRD event type.

ii. Ends with the timestamp recorded when the unit either presses the arrived button on their MCT or when the dispatcher presses the arrived button at their terminal if the status is communicated verbally. The 'cterm' column in the CAD table un_hi will denote whether the unit's or the dispatcher's terminal recorded the timestamp.
    - In the case where no arrived timestamp is recorded, the unit's "with patient" or transport timestamp may be used as a proxy to their arrival time.

iii. Outliers are any calls with a response time longer than three standard deviations above the average response time during the preceding 90 days. This represents approximately 0.3 percent of the dataset.

f. 90th Percentile

i. The value representing ninety percent of the data or the value that would be expected in nine out of ten instances of a phenomenon. For example, a turnout 90th percentile of 60 seconds indicates that on nine out of ten turnouts, units are enroute within 60 seconds of being dispatched.
    - Removing outliers prior to reporting the industry-standard ninetieth percentile, a more reliable reflection of reality is achieved. While the ninetieth percentile is not a measure of central tendency, it is a descriptor of the dataset's distribution and is thus vulnerable to gross outliers. Calculating the ninetieth percentile during the preceding 90 days also accounts for seasonal differences in response time that may result from phenomena such as extreme weather or significant changes in call volume due to population shifts.

5. Identifying Outliers

Data tables will be maintained and loaded daily in the FRD's data warehouse capturing each call dispatched in Fairfax County where the dispatched event type was categorized as an emergency medical service or fire emergency. During the process of loading this data, the outlier values for the preceding 90 days and for the entire dataset will be calculated. Columns denoting whether the real value is an outlier will display a 1 and a 0 if not. These data tables will feed a Power BI report which summarizes and trends the data over time.

Additional data tables will be maintained and loaded daily in the FRD's data warehouse capturing each call where an FRD unit was dispatched, both inside and outside Fairfax County, where the dispatched event type was categorized as an emergency medical service or fire emergency. During the process of loading this data, the outlier values for the preceding 90 days and for the entire dataset will be calculated. Columns denoting whether the real value is an outlier will display a 1 and a 0 if not. These data tables will feed a Power BI report which summarizes and trends the data over time. This dataset will allow for the isolation and analysis of calls requiring CAD2CAD transmissions which incur a processing delay.

Exclusions: Calls dispatched in Fairfax County as a public service where emergency response is not required. Included in these call types are Elevator Incident (ELEV), Lock Out (LOCKF), and PUBLIC SERVICE: LIFT ASSIST (LIFT), etc.

6. Handling Outliers

All datasets should be examined for outliers prior to analysis or reporting. Outliers may be identified through manual or programmatic exploration. Outliers should either be retained, transformed, or excluded.

For response times, outliers will be handled using one or more of the following methods, listed in order of application. In accordance with the Analysis Standards outlined in SOP X, the handling of any outliers should be documented in the methodology accompanying the analysis and/or its derivative product(s).

**Call processing**

a. *Retain* for response times analysis any outliers identified as True in *4.a.i*.
b. Continuous collaboration with DPSC to minimize the frequency with which erroneous times are recorded due to known issues impacting data quality and integrity.
    i. On a weekly basis, investigate high outliers to determine whether the call processing time was long due to extenuating circumstances of the call or due to human behavior or technical glitches which may be corrected. When appropriate, changes to the timestamps in the source data may be made to transform reflect true call processing.
c. *Exclude* from response times analysis any incident where the call processing time is deemed an outlier as defined in *0*.

**Turnout**

a. *Retain* for response times analysis any outliers identified as True in *4.a.i*.
b. Continuous messaging and education to the field regarding the expectation of apparatus personnel behavior in accurately documenting unit status.
c. Continuous collaboration with DPSC to minimize the frequency with which erroneous times are recorded due to known issues impacting data quality and integrity.

a. A Power BI report displaying calls with chronologically mis-ordered or missing timestamps during CAD downtime will be shared with DPSC. When appropriate, changes to the timestamps in the source data may be made to transform reflect true call processing.

d. *Exclude* from response times analysis any incident where the turnout time is deemed an outlier as defined in *0*.

7. Policy Review

At a minimum of twice annually, this policy should be reviewed by the DGC to determine its efficacy. Any additions or edits to the policy should be at the direction of the DGC, approved via memo through the Office of the Fire Chief, and announced to the FRD via Informational Bulletin.

Last Review Date

March X, 2025

Revision History

[i] Outlier Definition & Meaning - Merriam-Webster